Face

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Face

Boundaries

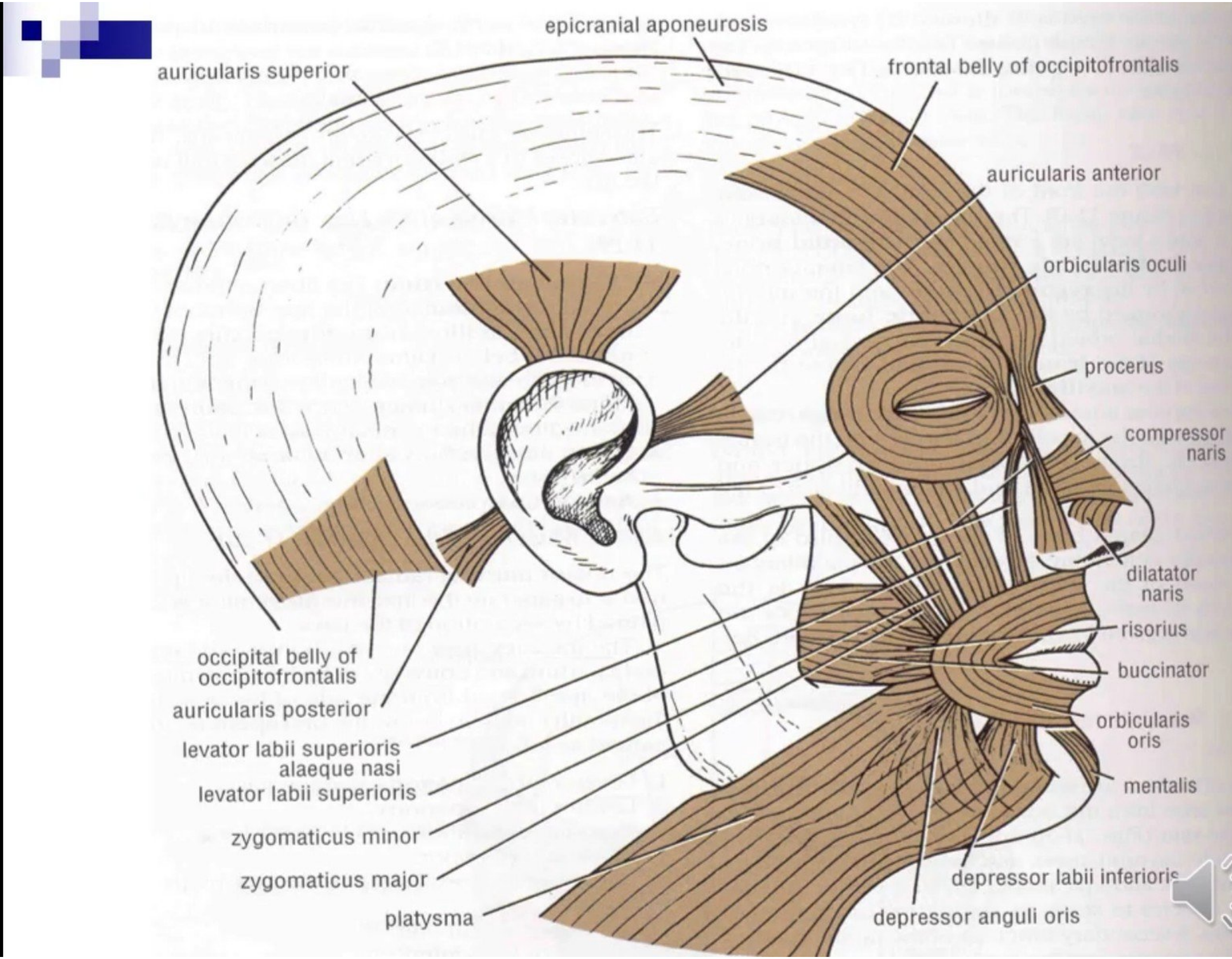
- Extends superiorly to the hair line, inferiorly to the chin and base of mandible, and on each side to auricle
- Forehead is common to both scalp and face

Skin

- Very vascular
- Due to rich vascularity face blush and blanch
- Wounds of face bleed profusely but heal rapidly
- Results of plastic surgery are excellent on face
- Facial skin is rich in sebaceous gland and sweat gland
- Sebaceous gland keep the skin oily but also cause acne in adult
- Sweat gland regulate body temperature

Facial muscle

- Called muscle of facial expression and lie in superficial fascia
- Embryologically they develop from mesoderm of 2^{nd} branchial arch, therefore supplied by facial nerve


epicranial aponeurosis
auricularis superior
frontal belly of occipitofrontalis
auricularis anterior
orbicularis oculi
procerus
compressor naris
dilatator naris
risorius
buccinator
orbicularis oris
mentalis
depressor labii inferioris
depressor anguli oris
platysma
zygomaticus major
zygomaticus minor
levator labii superioris
levator labii superioris alaeque nasi
auricularis posterior
occipital belly of occipitofrontalis

Orbicularis oculi

- 3 parts-
- Orbital part
  - Originate from medial part of medial palpebral ligament and form concentric rings, return to point of origin

  Action –closes the lids tightly
- Palpebral part
  - Originate from lateral part of medial palpebral ligament
  - Insert into lateral palpebral raphe

  Action-closes the lids gently
- Lacrimal part
  - Originate from lacrimal fascia& lacrimal bone
  - Insert into upper &lower tarsi

  Action-dilate lacrimal sac

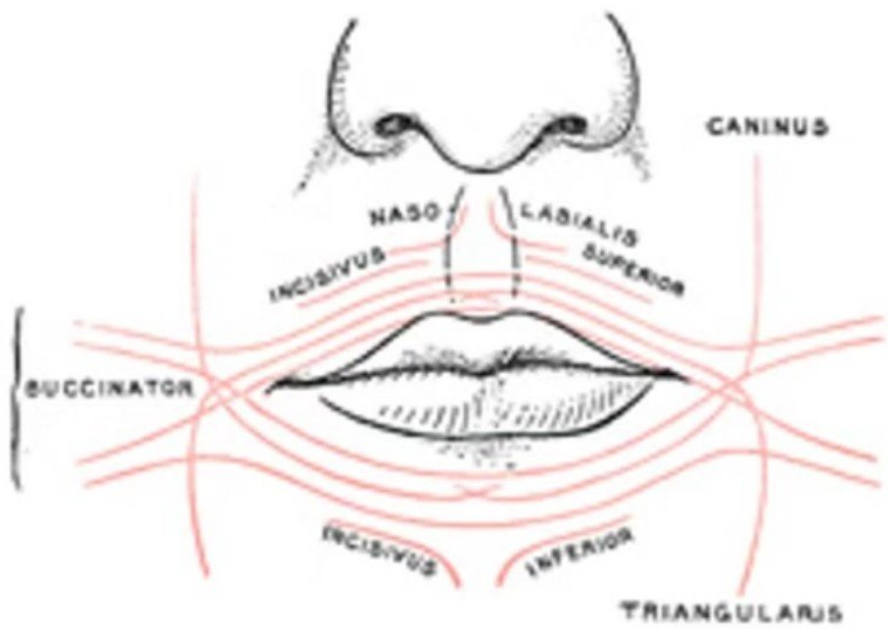
Orbicularis auris

- Originate from maxilla above incisor teeth and insert into skin of lip.

  Action –closes the mouth

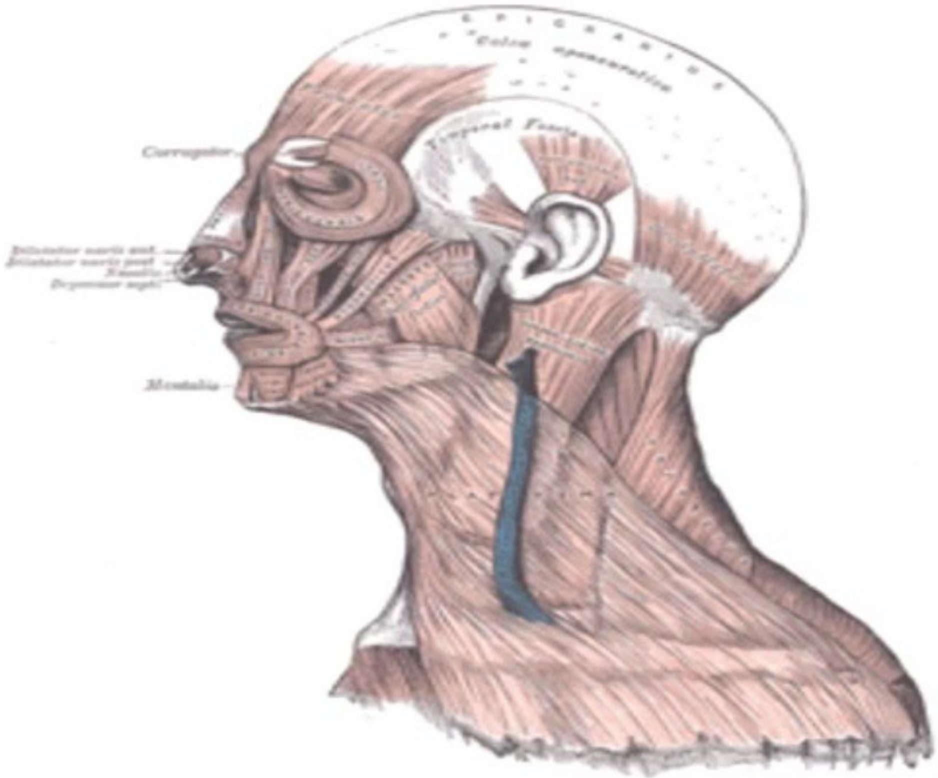
Buccinator

- Upper fibers
  - Origin- from maxilla opposite molar teeth
  - Insertion-upper lip
- Lower fibers
  - Origin-from mandible opposite molar teeth
  - Insertion-lower lip
- Middle fibers
  - Origin –from pterigomandibular raphe
  - Insertion-decussate before passing to lips
- Action- prevent accumulation of food in vestibule of mouth



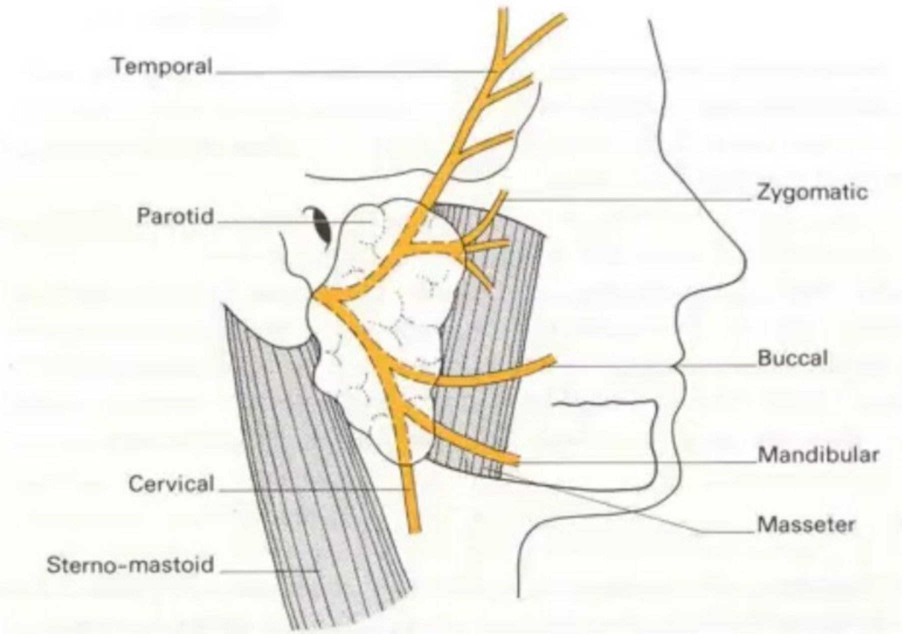
Platysma

- Origin– upper part of pectoral and deltoid fascia
- Insertion– base of mandible, skin of lower face and lip
- Action– releases pressure of skin on the subjacent veins, depress mandible, pulls angle of mouth downwards



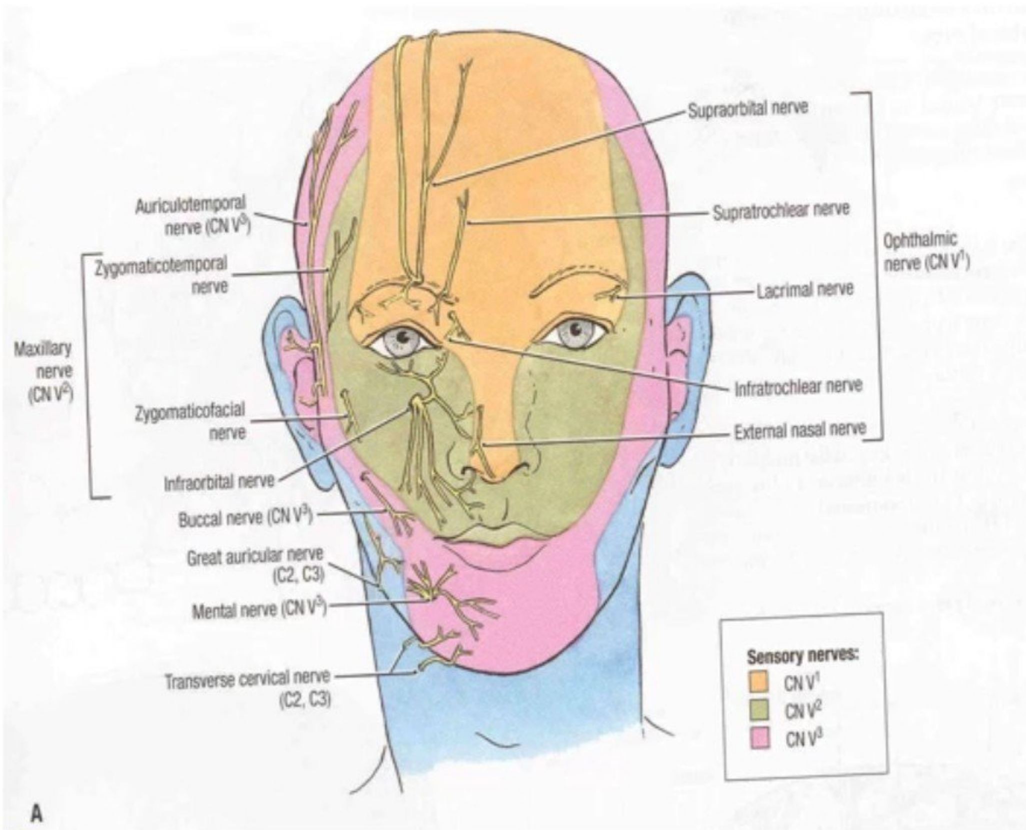
Nerve supply of face

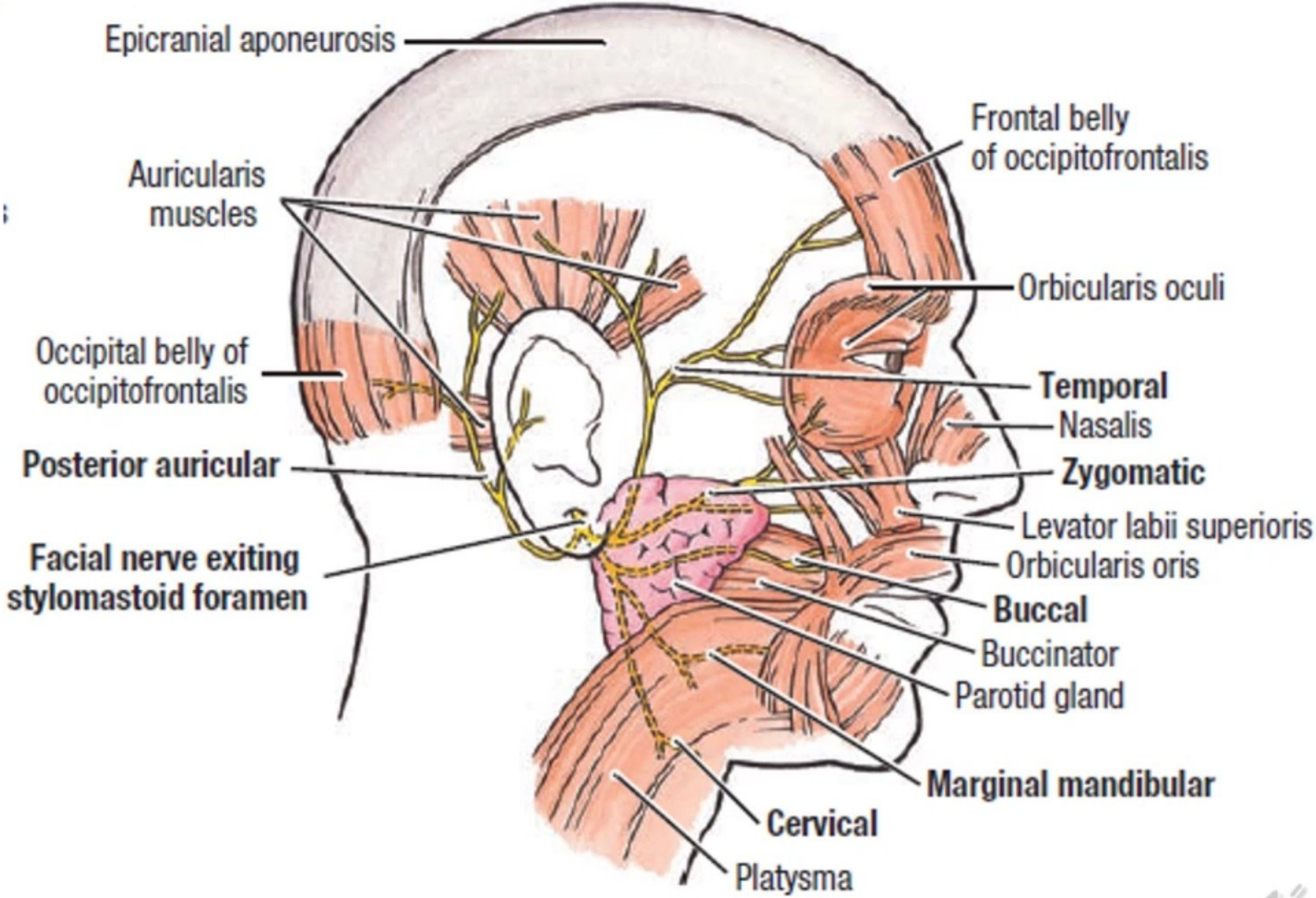
- Motor supply
  - Facial nerve



Sensory supply

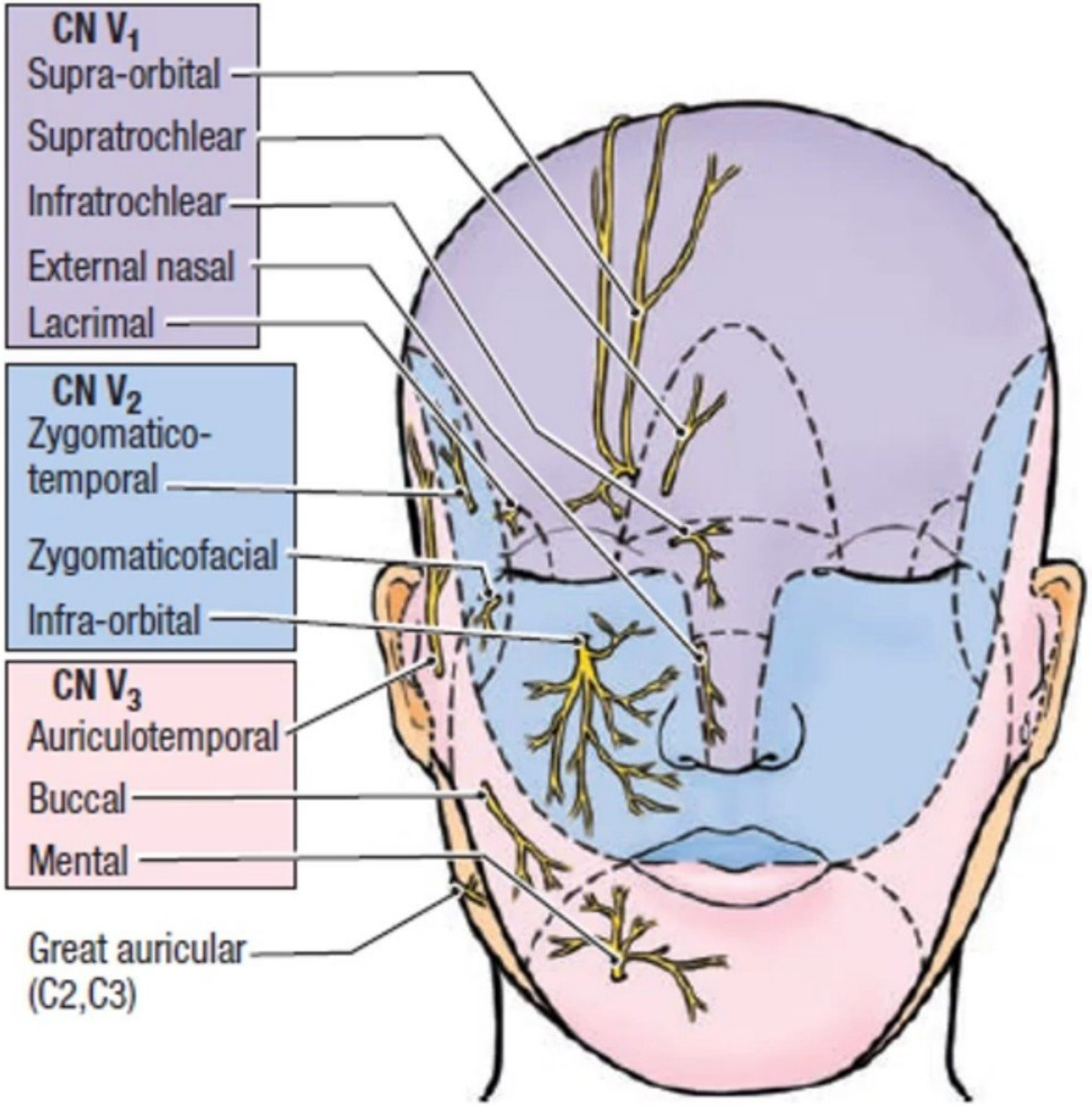
- Ophthalmic division
  - Supratrochlear
  - Supraorbital
  - Lacrimal
  - Infratrochlear
  - External nasal
- Maxillary nerve
  - Infraorbital
  - Zygomaticofacial and zygomaticotemporal
- Mandibular nerve
  - Auriculotemporal
  - Buccal nerve
  - Mental
- Skin over the mandibular angle is supplied by ant. Div. Of greater auricular n.



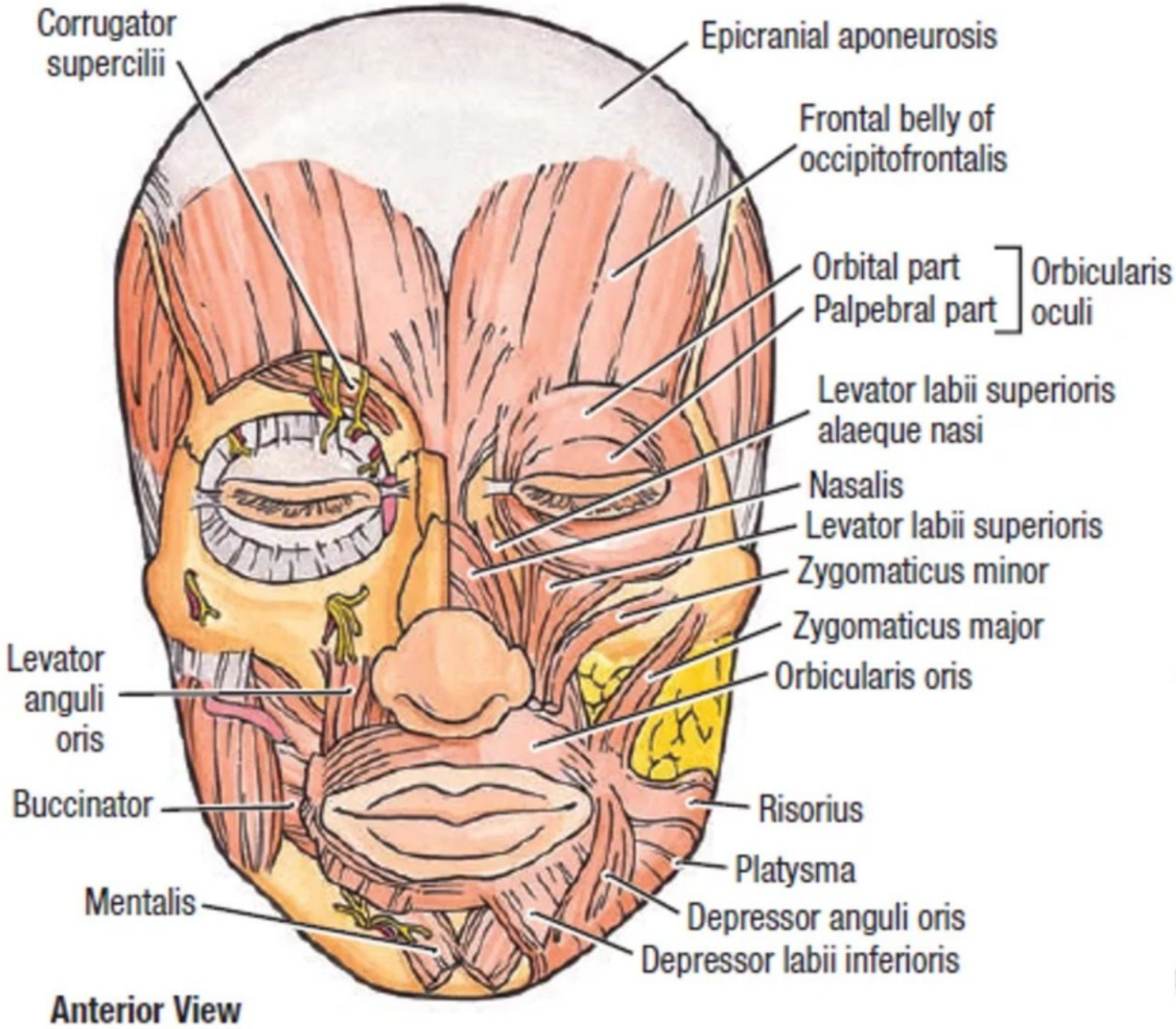


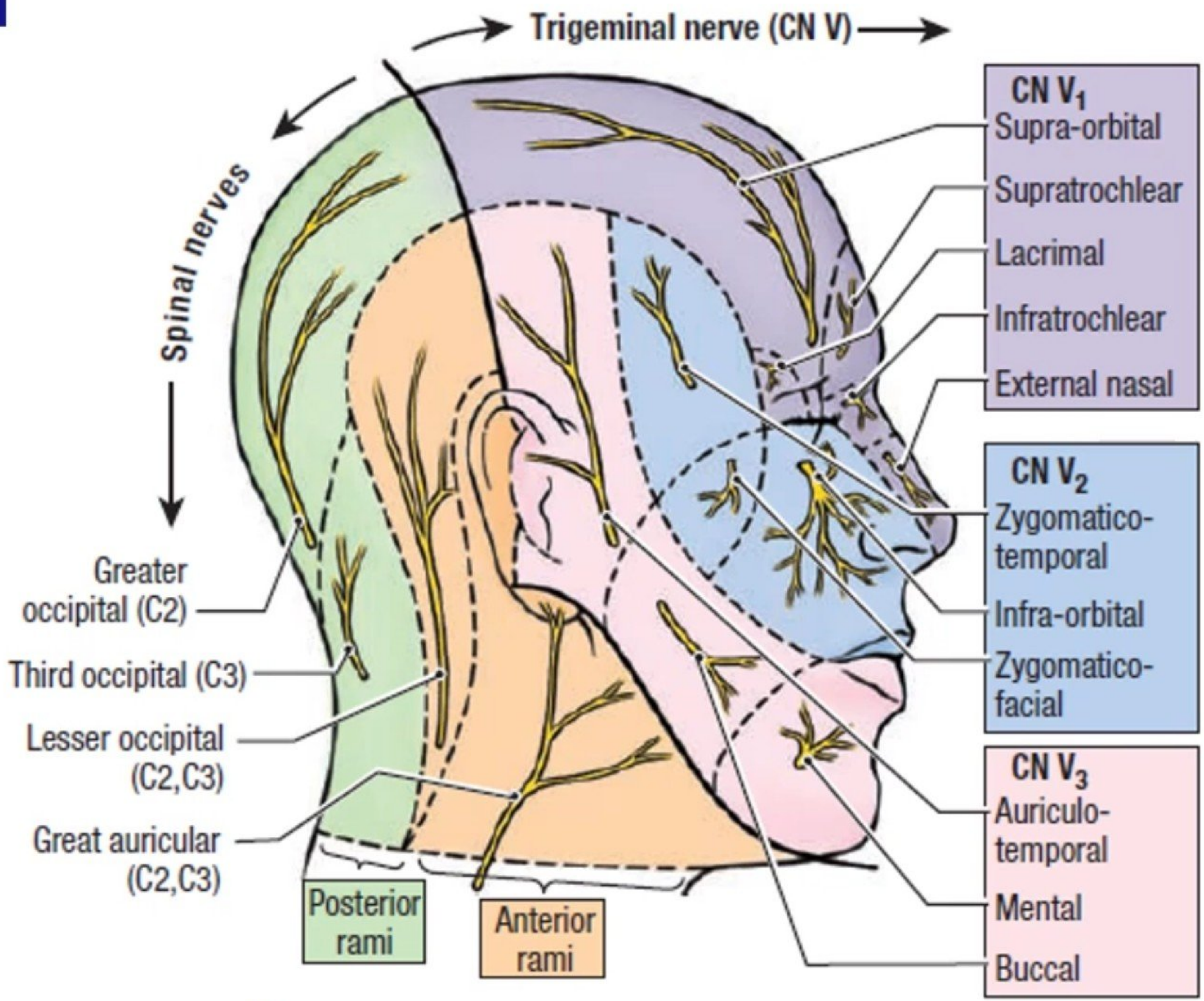
. Lateral View

Bold = Branches of facial nerve (motor)



A. Anterior view

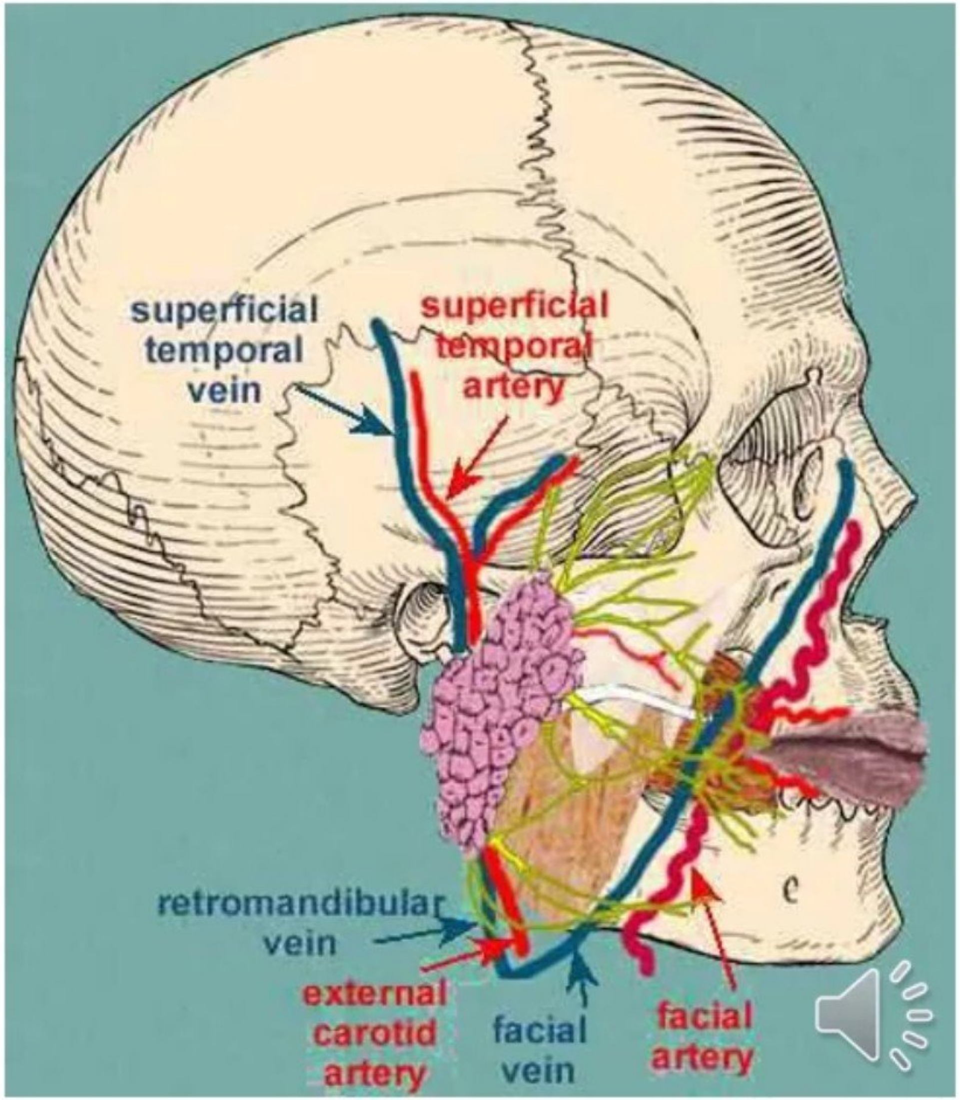

Corrugator supercilii
Epicranial aponeurosis
Frontal belly of occipitofrontalis
Orbital part
Palpebral part
Orbicularis oculi
Levator labii superioris alaeque nasi
Nasalis
Levator labii superioris
Zygomaticus minor
Zygomaticus major
Orbicularis oris
Levator anguli oris
Buccinator
Risorius
Platysma
Mentalis
Depressor anguli oris
Depressor labii inferioris
Anterior View

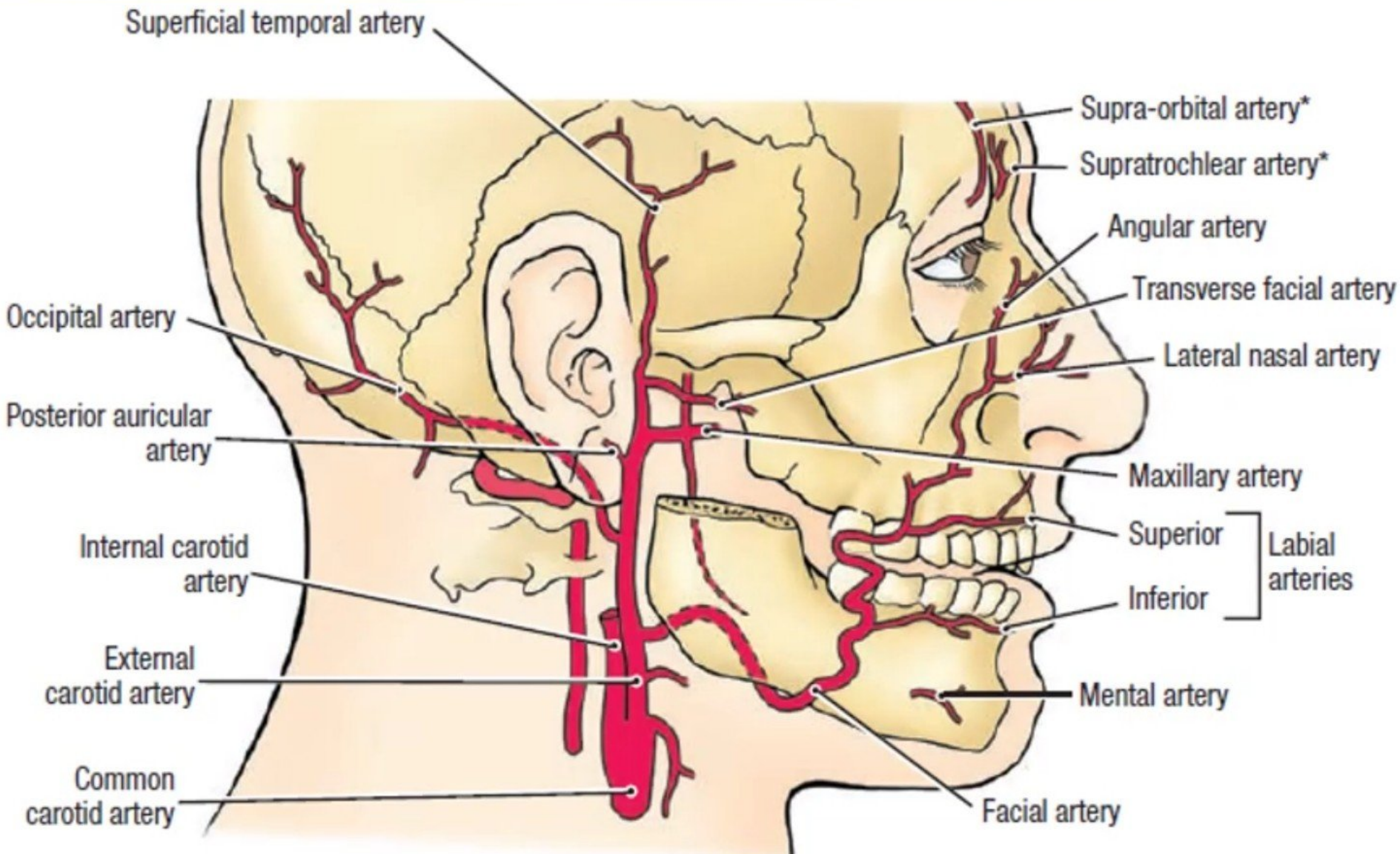


B. Lateral View (Cervical plexus)

Blood supply of face

- Arterial supply-
  - Facial artery
  - Superficial temporal artery
  - Ophthalmic artery
    - Supraorbital and
    - Supratrochlear



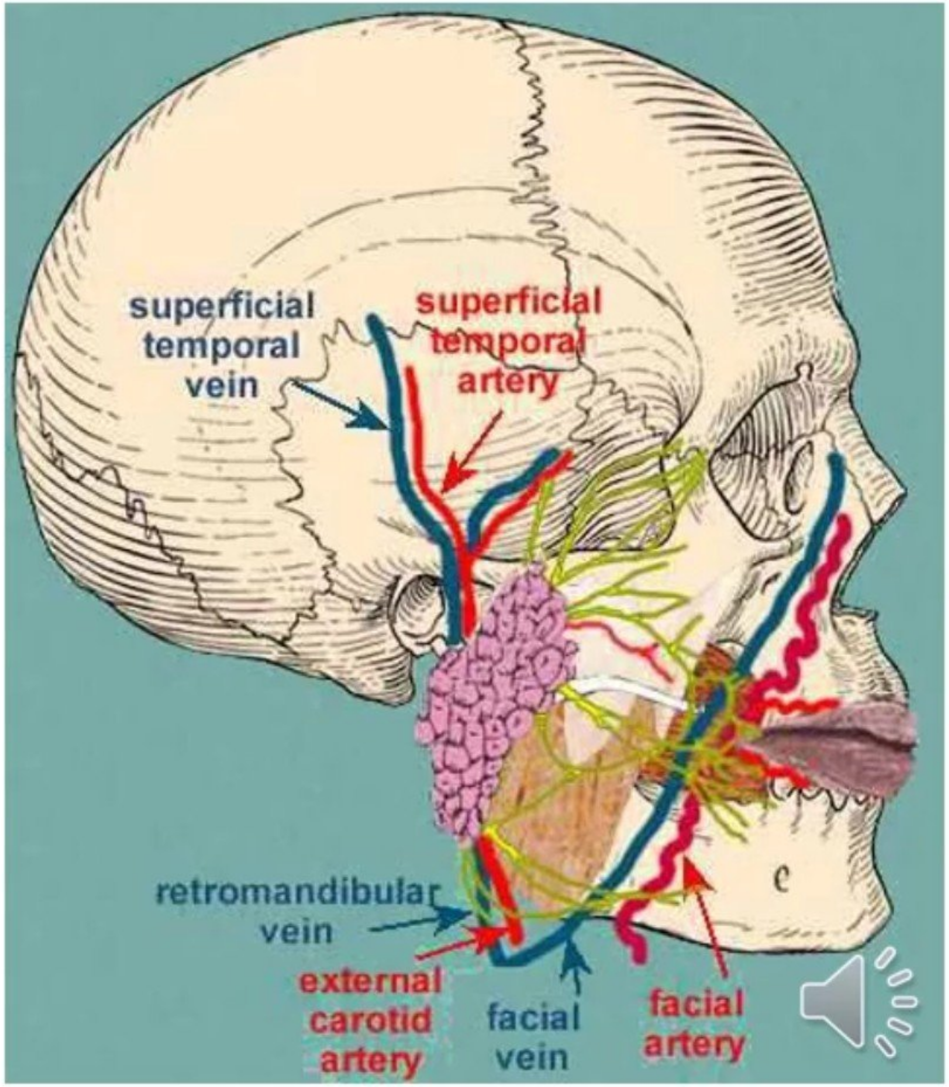


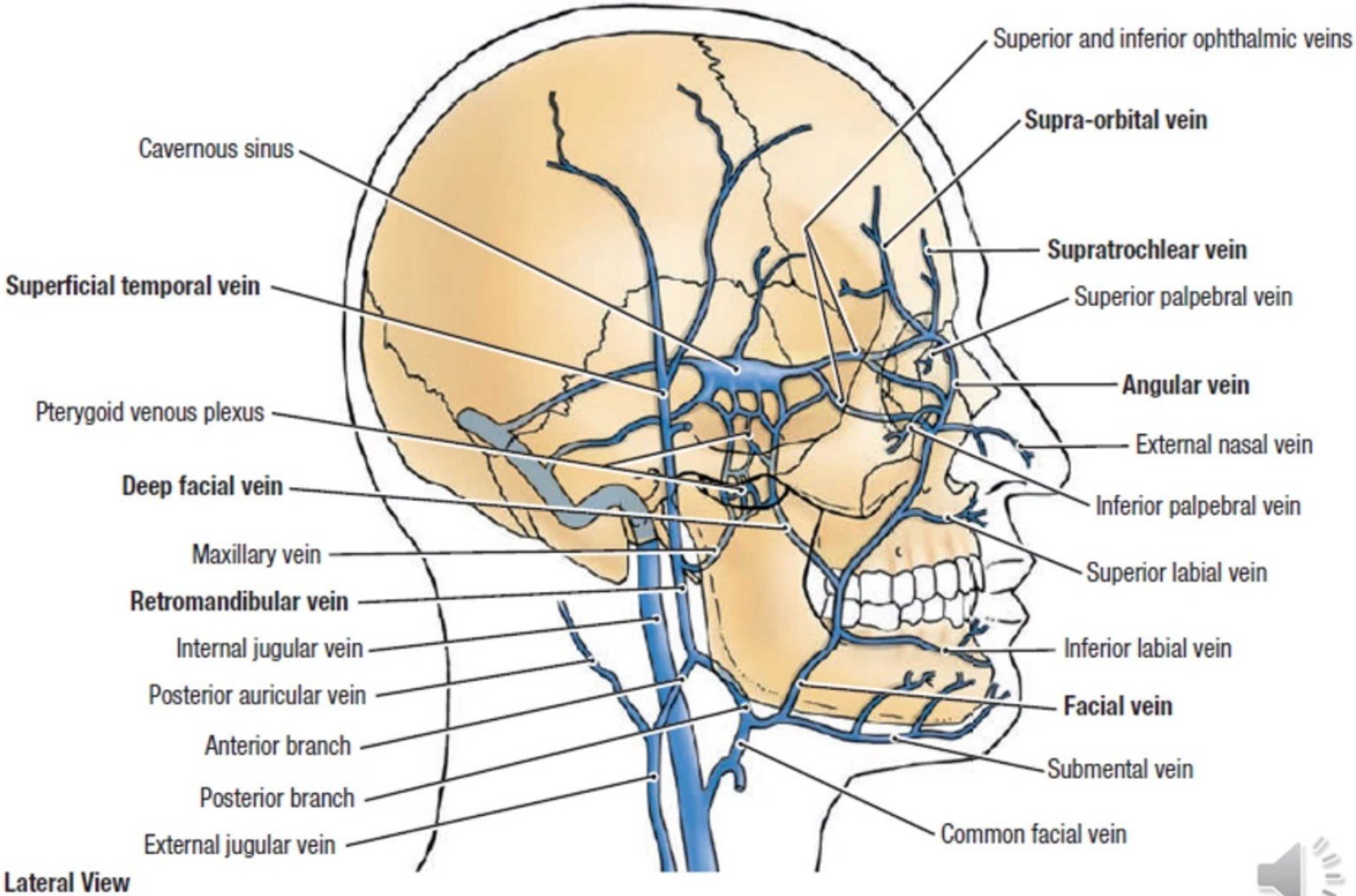
Lateral View

*Source= internal carotid artery (ophthalmic artery); all other labeled arteries are from external carotid

Venous drainage

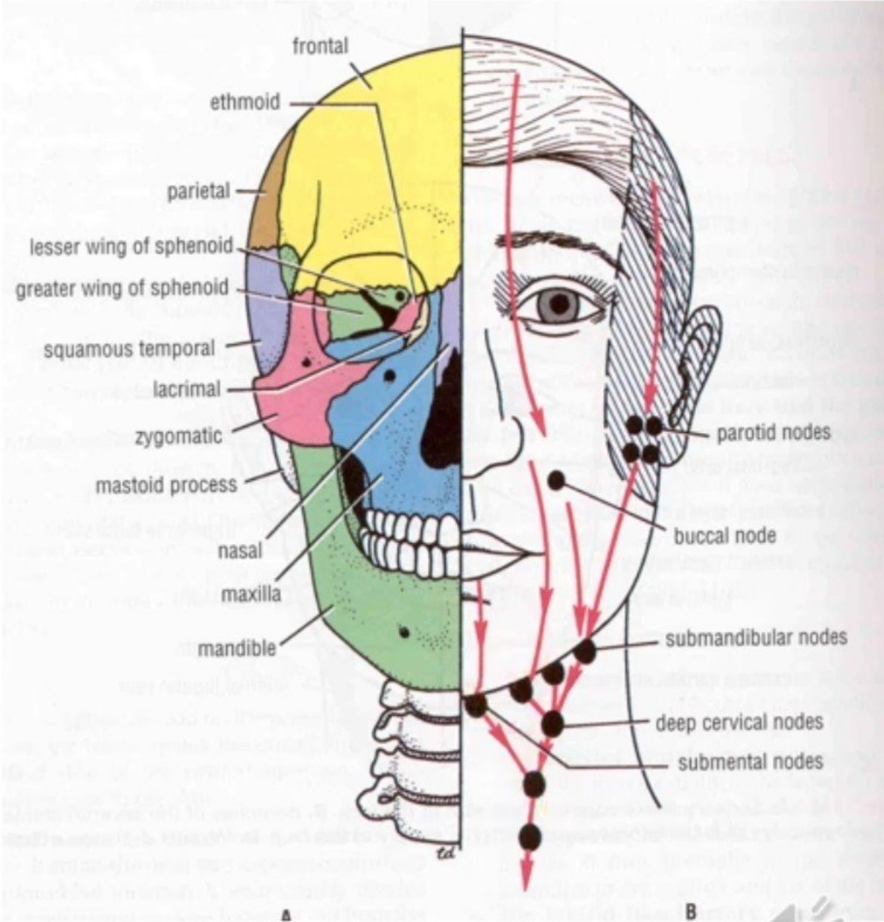
- Vein follow the arteries and drain into common facial vein and retromandibular vein
- Deep connections of facial vein-
- Communication between supraorbital &superior ophthalmic vein
- With pterigoid plexus of vein through deep facial vein.
- Superior ophthalmic vein & ptergoid plexus of vein communicate with cavernous sinus




Superior and inferior ophthalmic veins
Supra-orbital vein
Cavernous sinus
Supratrochlear vein
Superficial temporal vein
Superior palpebral vein
Angular vein
Pterygoid venous plexus
External nasal vein
Deep facial vein
Inferior palpebral vein
Maxillary vein
Superior labial vein
Retromandibular vein
Internal jugular vein
Inferior labial vein
Posterior auricular vein
Facial vein
Anterior branch
Submental vein
Posterior branch
External jugular vein
Common facial vein
Lateral View

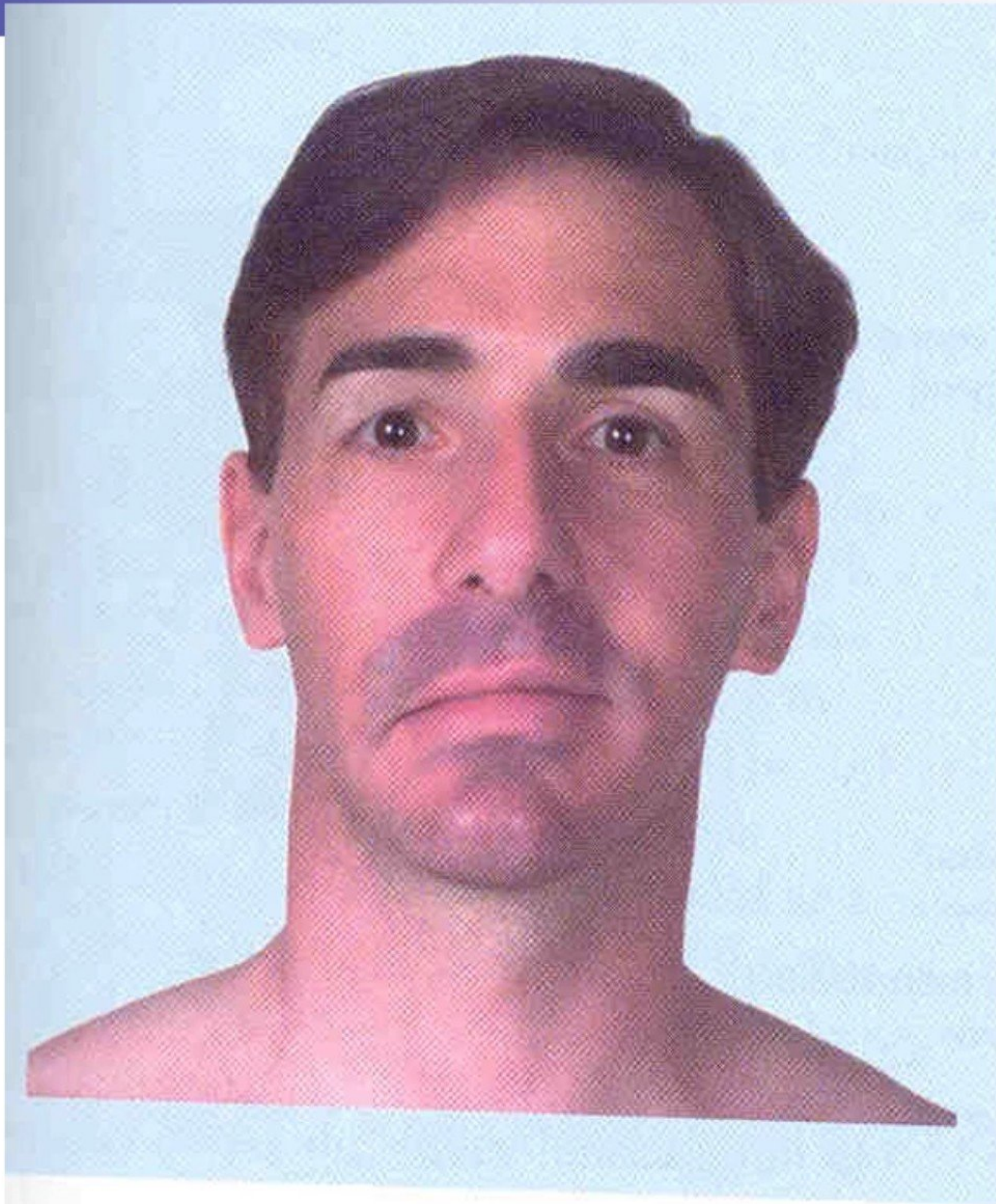
Lymphatic drainage

- 3 territories-
- Upper territories- greater part of forehead, lateral ½ of eye lid, conjunctiva, lateral part of cheek and parotid area– preauricular lymph node (parotid)
- Middle territories- median part of forehead, external nose, upper lip, lateral part of lower lip, medial ½ of eye lid, medial part of cheek, greater part of lower jaw– submandibular lymph node
- Lower territories- central part of lower lip, chin– sub mental lymph node

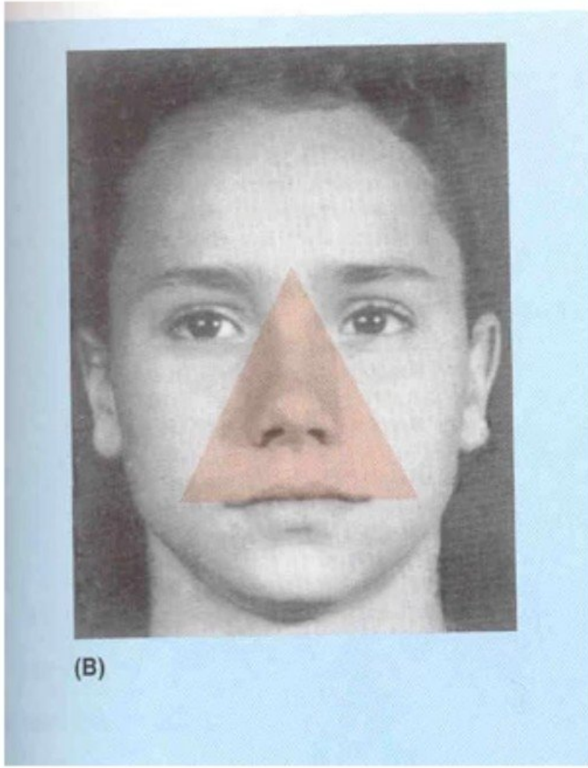


Applied

- Trigeminal neuralgia
  - Maxillary and mandibular nerve are involved
  - Excruciating pain in the region of distribution of these nerve
- In infranuclear lesions of facial nerve (eg, bell's palsy)- whole face is paralyzed
  - c/f
    - Affected side is motionless
    - Loss of wrinkles
    - Eye cannot be closed
    - In smiling the mouth is drawn to normal side
    - During mastication food accumulates in vestibule of mouth
- In supranuclear lesions of facial nerve only the lower part of face is paralyzed. The upper part (frontalis &part of orbicularis oculi) escapes due to its bilateral innervation



- Dangerous area of face- infections from face mainly from upper lip & nose can go to cavernous sinus through ophthalmic vein and deep facial vein


(B)

THANKS